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NORTH SYDNEY BOYS HIGH SCHOOL

2013 ASSESSMENT TASK 3

Preliminary Mathematics

General Instructions

- Working time – 50 minutes
 - Write on the lined paper in the booklet provided
 - Write using blue or black pen
 - Board approved calculators may be used
 - All necessary working should be shown in every question
 - Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

☐ Mr Lin

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	Total	Total %
Mark	$\frac{\quad}{13}$	$\frac{\quad}{14}$	$\frac{\quad}{16}$	$\frac{\quad}{43}$	$\frac{\quad}{100}$

Question 1 (13 marks)

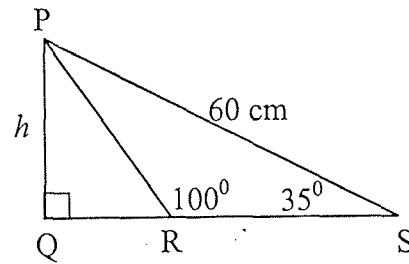
(a) Find the exact value of $\cos 300^\circ$

2

(b) If $\cos \alpha = \frac{2}{7}$ and $\sin \alpha < 0$ find the exact value of $\tan \alpha$ in the domain $0^\circ \leq \alpha \leq 360^\circ$

3

(c) In the diagram, $\angle PRS = 100^\circ$, $\angle PSR = 35^\circ$, $PS = 60\text{cm}$, $\angle PQR = 90^\circ$ and $PQ = h\text{ cm}$.



(i) Show that $PR = \frac{60 \sin 35^\circ}{\sin 100^\circ}$

1

(ii) Hence find h correct to 2 decimal places.

2

(d) Graph the region of the intersection given by the inequations:

$$y \geq 0, x \geq 0 \text{ and } y < 4 - x^2$$

3

(e) Find the exact area of an equilateral triangle with side length of $x\text{cm}$.

2

Question 2 (14 marks)

(a) Show that $\frac{1}{1-\sin \theta} + \frac{1}{1+\sin \theta} = \frac{2 \sec^2 \theta}{\sec^2 \theta}$

3

(b) Solve $4 \cos^2 \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$

3

(c) Show that the line $3x + 4y + 10 = 0$ is a tangent to the circle:

$$(x - 2)^2 + (y + 3)^2 = \frac{16}{25}$$

3

(d) A plane travels 1380km from A to B on a bearing of 58° T. The plane then travels 985km on a bearing of 135° T to a point C.

(i) Illustrate this information on a sketch and find the distance of CA, correct to one decimal place.

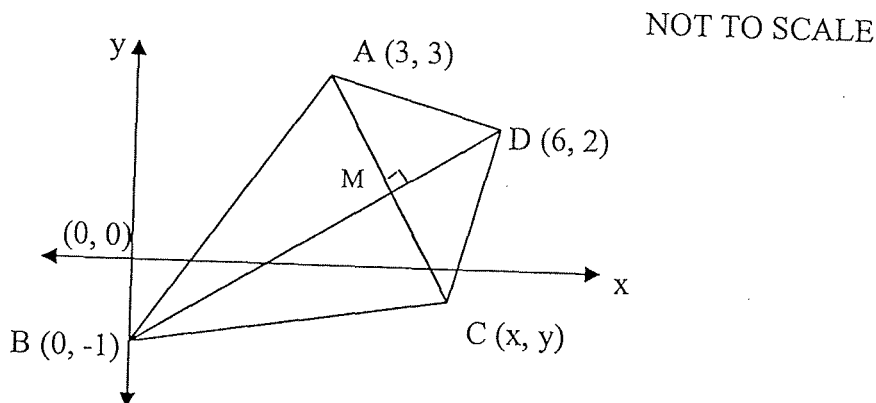
3

(ii) Find the bearing of A from C to the nearest minute.

2

Question 3 (16 marks)

In the quadrilateral ABCD, the points A, B, C and D are (3,3), (0,-1) and (6,2) respectively. The line BD bisects the line AC at right angles at the point M.



(a) Find the distance BD.

2

(b) Show that the gradient of BD is $\frac{1}{2}$.

2

(c) Show that the equation of the line BD is $x - 2y - 2 = 0$.

2

(d) Show that the equation of AC is $2x + y - 9 = 0$.

2

(e) Find the coordinates of M.

3

(f) Show that the coordinates of C are (5, -1).

2

(g) Find the area of ABCD.

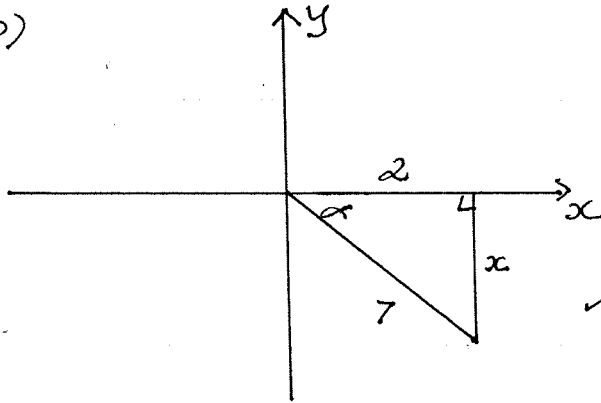
3

End of paper

Question 1

$$\begin{aligned}
 (a) \cos 300^\circ &= \cos (360^\circ - 60^\circ) \\
 &= \cos 60^\circ \\
 &= \frac{1}{2}
 \end{aligned}$$

(b)



$$x^2 + 2^2 = 7^2$$

$$x^2 = 45$$

$$x = \sqrt{45}$$

$$\therefore \tan \alpha = -\frac{3\sqrt{5}}{2} \quad (\tan \alpha < 0)$$

(c) (i) Using sine rule in $\triangle PRS$,

$$\frac{PR}{\sin 35^\circ} = \frac{60}{\sin 100^\circ}$$

$$PR = \frac{60 \sin 35^\circ}{\sin 100^\circ}$$

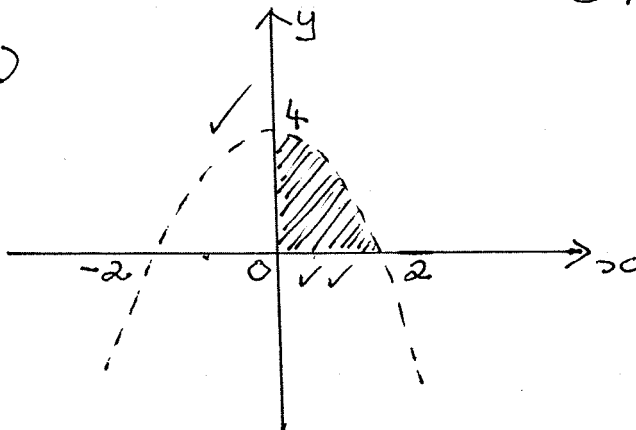
$$(ii) \angle PQR = 180^\circ - 100^\circ = 80^\circ$$

$$\text{In } \triangle PQR, \quad h = \frac{PR \sin 80^\circ}{PR}$$

$$h = PR \sin 80^\circ = \frac{60 \sin 35^\circ \sin 80^\circ}{\sin 100^\circ}$$

$$= 34.41 \quad (2 \text{ d.p.})$$

(d)



$$\begin{aligned}
 (e) A &= \frac{1}{2} ab \sin C \\
 &= \frac{1}{2} \times x \times x \times \sin 60^\circ \\
 &= \frac{x^2}{2} \times \frac{\sqrt{3}}{2} \\
 &= \frac{\sqrt{3} x^2}{4} \text{ cm}^2
 \end{aligned}$$

Question 2

(a) Show $\frac{1}{1-\sin\theta} + \frac{1}{1+\sin\theta} = 2\sec^2\theta$

$$\text{LHS} = \frac{1+\sin\theta}{(1-\sin\theta)(1+\sin\theta)} + \frac{1-\sin\theta}{(1-\sin\theta)(1+\sin\theta)} \quad \checkmark$$

$$= \frac{2}{1-\sin^2\theta}$$

$$= \frac{2}{\cos^2\theta} \quad \checkmark$$

$$= 2\sec^2\theta \quad \checkmark$$

$$= \text{RHS}$$

(b) $4\cos^2\theta = 1$

$$\cos^2\theta = \frac{1}{4} \quad \checkmark$$

$$\cos\theta = \pm\frac{1}{2} \quad \checkmark$$

$$\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ \quad \checkmark$$

(c) $(x-2)^2 + (y+3)^2 = \frac{16}{25}$

Centre $(2, -3)$ $r = \frac{4}{5}$

Perpendicular distance from centre to line:

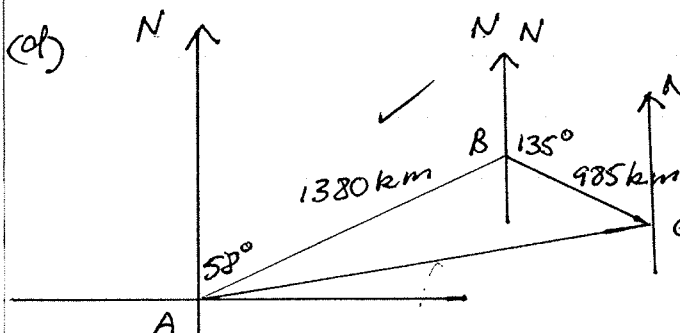
$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \quad \checkmark$$

$$= \frac{|3 \times 2 + 4 \times -3 + 10|}{\sqrt{9 + 16}} \quad \checkmark$$

$$= \frac{4}{5} \quad \checkmark$$

$$= r$$

$\therefore$ the line is a tangent to the circle



(i) $\angle ABC = 58^\circ + (180^\circ - 135^\circ) = 103^\circ$

$$\checkmark CA^2 = 1380^2 + 985^2 - 2 \cdot 1380 \cdot 985 \cos 103^\circ$$

$$\checkmark CA = 1867.1 \text{ km (1 d.p.)}$$

(ii) $\sin \angle BCA = \frac{1380 \sin 103^\circ}{1867.1}$ $\checkmark$

$$\angle BCA = 46^\circ 4'$$

$$\checkmark \therefore \text{Bearing of A from C} = 360^\circ - (46^\circ 4' + 45^\circ) = 268^\circ 56'$$

Question 3

$$\begin{aligned} \text{(a)} \quad d &= \sqrt{(2+1)^2 + (6-0)^2} \\ &= \sqrt{45} \\ &= 3\sqrt{5} \text{ units} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad m_{BD} &= \frac{2+1}{6-0} \\ &= \frac{1}{2} \end{aligned}$$

$$\text{(c)} \quad y = mx + b$$

$$BD: y = \frac{1}{2}x - 1$$

$$2y = x - 2$$

$$\therefore x - 2y - 2 = 0$$

$$\text{(d)} \quad m_{AC} = -2$$

$$\begin{aligned} y - 3 &= -2(x - 3) \\ &= -2x + 6 \end{aligned}$$

$$\therefore 2x + y - 9 = 0$$

$$\begin{aligned} \text{(e)} \quad x - 2y - 2 &= 0 & \text{--- (1)} \\ 2x + y - 9 &= 0 & \text{--- (2)} \end{aligned}$$

$$\text{(2) - (1)} \quad 5y - 5 = 0$$

$$y = 1$$

$$x = 4$$

$$\therefore M(4, 1)$$

$$\text{(f)} \quad 4 = \frac{x+3}{2}$$

$$1 = \frac{y+3}{2}$$

$$x = 5$$

$$y = -1$$

$$\therefore C(5, -1)$$

$$\begin{aligned} \text{(g)} \quad A &= \frac{1}{2} BD \cdot AC \\ &= \frac{1}{2} \times 3\sqrt{5} \times 2\sqrt{5} \\ &= 15 \text{ unit}^2 \end{aligned}$$

$$\begin{aligned} * AC &= \sqrt{(-1-3)^2 + (5-3)^2} \\ &= 2\sqrt{5} \end{aligned}$$